Tut quiz question bank -5

Q1. Write the truth table for an SR NOR latch. An SR NOR latch initially has Q = 0. For the input sequence (S, R) = (1,0), (0,0), (0,1), (0,0), (1,0), (0,1), (0,0), determine the value of Q after each input combination.

Q2. Write the truth table for an SR NOR latch. An SR NOR latch is required to produce the output sequence Q: 0 -> 1 -> 1 -> 0 -> 0 -> 1. Give one possible sequence of S and R inputs that produces this output.

Q3. An SR NOR latch has inputs S (Set) and R (Reset). Modify the latch to function as a D latch using a data input D and an enable input E. Design the logic required to generate S and R, and draw the resulting circuit.

Q4. Design a D latch using a 2:1 multiplexer. Clearly indicate the inputs of the multiplexer, the select input, and the feedback connection required to obtain the D-latch behavior.

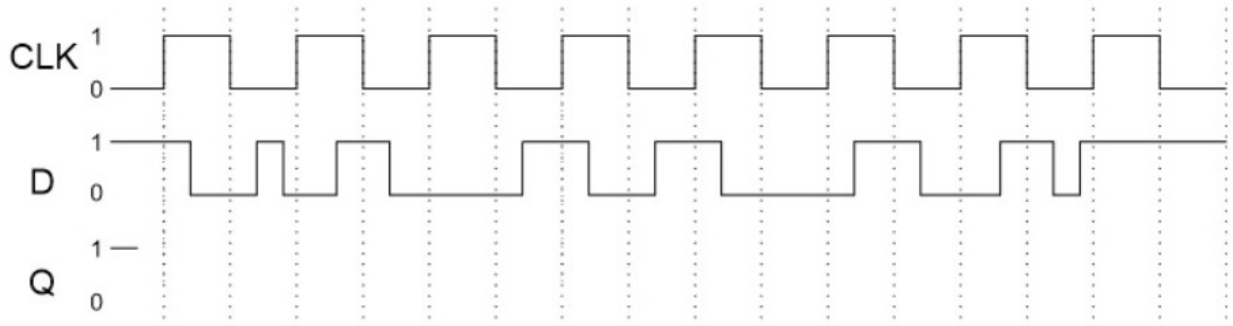
Q5. Distinguish between level-triggered and edge-triggered operation with respect to the control/clock signal. For an active-high D latch and a positive-edge-triggered D flip-flop, state when the output Q can change and illustrate the difference using suitable waveforms.

Q6. Construct a negative-edge-triggered (falling-edge-triggered) D flip-flop using appropriate logic elements and write its truth table.

Q7. Give the block diagram representation of a master-slave D flip-flop. Write its truth table and characteristic table, and derive the characteristic equation.

Q8. Give the block diagram representation of a master-slave D flip-flop. Write its truth table and excitation table, and derive the excitation equation.

Q9. The CLK and D waveforms shown in the figure are applied to a rising-edge-triggered D flip-flop. If the initial value of Q is 1, draw the resulting Q waveform. Assume gate delays are negligible.



Q10. A positive-edge-triggered D flip-flop is connected such that its output Q is fed back to its D input through an inverter. If the initial value of Q is 0, determine the value of Q after each of the first four rising edges of the clock. Draw the corresponding timing waveform.